Bloch-Wigner sequence and linear groups over finite rings

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Abstract

The Bloch-Wigner exact sequence is an important result in K-theory that connects the third homology group of $\mathrm{SL}_2(\mathbb{C})$ with the Bloch group, which encodes information about scissors congruences of polyhedra in hyperbolic geometry. These concepts provide deep insights into the interplay between algebra, geometry, and topology.

Hutchinson, Coronado, Mirzaii and Pérez have extended the Bloch-Wigner sequence to local rings [1, 2]. In this work, we apply these results to study the low-dimensional homology groups of SL_2 over finite commutative rings. We also discuss other achievements in the area that have been obtained using these methods.

The complex of unimodular vectors

Let A be a commutative ring. Let $\mathrm{D}_n(A)$ be the subgroup of $\mathrm{GL}_n(A)$ generated by diagonal matrices. Let $\mathrm{GE}_n(A)$ be the subgroup of $\mathrm{GL}_n(A)$ generated by $\mathrm{D}_n(A)$ and $\mathrm{E}_n(A)$. A ring A is called a GE_n**-ring** if

$$\mathrm{GE}_n(A) = \mathrm{GL}_n(A).$$

A ring A is called a GE**-ring**, if it is a GE_n-ring for any n

Proposition 1. *Semilocal rings and Euclidean domains are GE-rings. In particular, finite rings are GE-rings.*

A vector $\begin{pmatrix} u_1 \\ u_2 \end{pmatrix} \in A^2$ is said to be *unimodular* if there exists a vector $\begin{pmatrix} v_1 \\ v_2 \end{pmatrix}$ such that $\begin{pmatrix} u_1 & v_1 \\ u_2 & v_2 \end{pmatrix} \in \mathrm{GL}_2(A)$. For any $v \in A^2 - \{0\}$, let $\langle v \rangle := \{av : a \in A\}$.

Let $X_n(A^2)$ be the free abelian group generated by the set of $(n+1)$-tuples $(\langle v_0 \rangle, \ldots, \langle v_n \rangle)$, such that every $v_i \in A^2$ is unimodular and $(v_i, v_j) \in \mathrm{GL}_2(A)$ for $i \neq j$.
$X_n(A^2)$ is as a left $\mathrm{GL}_2(A)$-module by the coordinate-wise action. Note that $\mathrm{SL}_2(A)$ acts transitively on the generators of $X_0(A^2)$ and $X_1(A^2)$.

Let us define the l-th differential operator $\partial_l : X_l(A^2) \to X_{l-1}(A^2), \ \ l \geq 1$, as an alternating sum of face operators. Let ∂_0 be the augmentation map. Hence we have the complex

$$X_\bullet(A^2) \to \mathbb{Z} : \ \cdots \longrightarrow X_2(A^2) \xrightarrow{\partial_2} X_1(A^2) \xrightarrow{\partial_1} X_0(A^2) \to \mathbb{Z} \to 0.$$

Theorem 1 (Hutchinson)**.** *Let A be a commutative ring.*
(i) *$X_\bullet(A^2) \to \mathbb{Z}$ is exact in dimension < 1 if and only if A is a GE_2-ring.*
(ii) *If A is local, then $X_\bullet(A^2) \to \mathbb{Z}$ is exact in dimension < 2.*
(iii) *If A is local, then $X_\bullet(A^2) \to \mathbb{Z}$ is exact in dimension $i < |A/\mathfrak{m}_A|$.*

We denote

$$\mathrm{B}(A) = \left\{ \begin{pmatrix} a & b \\ 0 & a^{-1} \end{pmatrix} : a \in A^\times, b \in A \right\}, \quad \mathrm{T}(A) = \left\{ \begin{pmatrix} a & 0 \\ 0 & a^{-1} \end{pmatrix} : a \in A^\times \right\}.$$

The homologies of these matrices groups are related to the spectral sequence 1 which computes $H_*(\mathrm{SL}_2(A), \mathbb{Z})$. The map $\mathrm{B}(A) \to \mathrm{T}(A)$, given by $\begin{pmatrix} a & b \\ 0 & a^{-1} \end{pmatrix} \mapsto \begin{pmatrix} a & 0 \\ 0 & a^{-1} \end{pmatrix}$, induces the split extension of abelian groups

$$1 \to \mathrm{N}(A) \to \mathrm{B}(A) \to \mathrm{T}(A) \to 1,$$

where $\mathrm{T}(A) \simeq A^\times$ and $\mathrm{N}(A) \simeq A$.

Motivated by the classical Bloch-Wigner sequence, we consider the square class group

$$\mathcal{G}_A := A^\times/(A^\times)^2.$$

The element of $\mathcal{G}_A$ represented by $a \in A^\times$ is denoted by $\langle a \rangle$. We set $\langle\langle a \rangle\rangle := \langle a \rangle - 1 \in \mathcal{G}_A$. Let $\mathcal{W}_A$ be the set of $a \in A^\times$such that $1 - a \in A^\times$. Thus

$$\mathcal{W}_A := \left\{ a \in A : a(1-a) \in A^\times \right\}.$$

Let A be a GE_2-ring. Consider the exact sequence $0 \to Z_1(A^2) \overset{\mathrm{inc}}{\to} X_1(A^2) \overset{\partial_1}{\to} X_0(A^2) \to \mathbb{Z} \to 0$. Removing the first term and tensoring by the bar resolution over $\mathrm{SL}_2(A)$, we obtain the double complex

$$D_{\bullet,\bullet} : 0 \to F_\bullet \otimes_{\mathrm{SL}_2(A)} Z_1(A^2) \xrightarrow{\mathrm{id}_{F_\bullet} \otimes \mathrm{inc}} F_\bullet \otimes_{\mathrm{SL}_2(A)} X_1(A^2) \xrightarrow{\mathrm{id}_{F_\bullet} \otimes \partial_1} F_\bullet \otimes_{\mathrm{SL}_2(A)} X_0(A^2) \to 0,$$

This gives us the first quadrant spectral sequence

$$E^1_{p,q} = \begin{cases} H_q(\mathrm{SL}_2(A), X_p(A^2)) & \text{if } p = 0, 1 \\ H_q(\mathrm{SL}_2(A), Z_1(A^2)) & \text{if } p = 2 \\ 0 & \text{if } p > 2 \end{cases} \implies H_{p+q}(\mathrm{SL}_2(A), \mathbb{Z}). \tag{1}$$

After studying the lower terms of the spectral sequence 1, we obtain

Theorem 2. *For any GE_2-ring A we have the exact sequence*

$$H_2(\mathrm{B}(A), \mathbb{Z}) \twoheadrightarrow H_2(\mathrm{SL}_2(A), \mathbb{Z}) \to I(A) \xrightarrow{d^2_{2,0}} \mathcal{G}_A \oplus A_{A^\times} \to H_1(\mathrm{SL}_2(A), \mathbb{Z}) \to 0.$$

The refined Bloch group

From now on, assume that $|k| \geq 4$ and that $H_2(\mathrm{B}(A), \mathbb{Z}) = H_2(\mathrm{T}(A), \mathbb{Z})$.

Let $\overline{\mathcal{RP}}(A)$ be the quotient of the free $\mathcal{G}_A$-module generated by the symbols $[x], x \in \mathcal{W}_A$ over the subgroup generated by the elements

$$[x] - [y] + \langle x \rangle \left[\frac{y}{x}\right] - \left\langle x^{-1} - 1 \right\rangle \left[\frac{1 - x^{-1}}{1 - y^{-1}}\right] + \langle 1 - x \rangle \left[\frac{1-x}{1-y}\right], \quad x, y, x/y \in \mathcal{W}_A.$$

It is called the *refined scissors congruence group*. Consider the map $\lambda : \overline{\mathcal{RP}}(A) \to \mathbb{Z}[\mathcal{G}_A]$ given by $[x] \mapsto -\langle\langle x \rangle\rangle \langle\langle 1 - x \rangle\rangle$. Let $\overline{\mathcal{RP}}_1(A)$ be the kernel of this map.

Theorem 1. For any GE_2-ring A with the above properties, we have the commutative diagram

$$\begin{array}{ccc} \mathcal{RP}_1(A) & \xrightarrow{\lambda_1} & \dfrac{A^\times \wedge A^\times}{A^\times \wedge \mu_2(A)} \\ \downarrow & & \downarrow \\ \mathcal{P}(A) & \longrightarrow & (A^\times \wedge A^\times) \oplus S^2_{\mathbb{Z}}(A) \end{array}$$

where $S^2_{\mathbb{Z}}(A) := \left(A^\times \otimes A^\times\right) / \left\langle a \otimes b + b \otimes a : a, b \in A^\times \right\rangle$. Moreover, the vertical map on the right is given by $a \wedge b \ \mapsto \ (2a \wedge b, 2(a \otimes b))$ and the bottom horizontal map is given by $[a] \mapsto (a \wedge (1-a), -a \otimes (1-a))$.

The kernel of the map λ_1 is called the *refined Bloch group* and is denoted by $\mathcal{RB}(A)$.

Applications to finite rings

Finite commutative rings are Noetherian and Artinian, then products of local rings. So we focus on commutative finite local rings, which have order a power of a prime.

Let A be a finite commutative local ring. It will be important to understand the structure of the group of units of A, denoted by $A^\times$.

Proposition 1. Let A be a local ring with maximal ideal $\mathfrak{m}_A$ and let $k := A/\mathfrak{m}_A$ with $\mathrm{char} k = p$. Then $1 + \mathfrak{m}_A$ is a p-group in $A^\times$. Moreover $k^\times \simeq A^\times/(1 + \mathfrak{m}_A)$ and

$$A^\times \simeq k^\times \oplus (1 + \mathfrak{m}_A).$$

Corollary 1. *If $\mathrm{char}(k)$ is odd, then $\mathcal{G}_A \simeq \mathcal{G}_k \simeq \mathbb{Z}/2$.*

In this part we compute $H_2(\mathrm{SL}_2(A), \mathbb{Z})$ for finite local rings A.

Proposition 2. *By the Künneth formula, $H_2(A^\times, \mathbb{Z}) \simeq H_2(1 + \mathfrak{m}_A, \mathbb{Z})$. If $H_2(B(A)) = H_2(T(A))$, which happens for finite values of $|k|$, then*

$$H_2(B(A), \mathbb{Z}) = H_2(1 + \mathfrak{m}_A, \mathbb{Z}).$$

The augmentation ideal of $\mathbb{Z}[\mathcal{G}_A]$ is $\mathcal{I}(A) = \mathbb{Z}[\overline{-1} - \overline{1}]$. Let $a \in \mathcal{W}_A$, since $\langle a \rangle \in \mathbb{Z}_2$, it is easy to see that

$$\langle\langle a \rangle\rangle \langle\langle 1 - a \rangle\rangle = \begin{cases} 2(\overline{1} - \overline{-1}) & \text{if } \langle x \rangle = \langle 1 - x \rangle = \overline{-1} \\ 0 & \text{otherwise} \end{cases}$$

It can be shown that $d^2_{20} : \mathbb{Z}_2 \to \mathbb{Z}_2$ is the identity, so its kernel vanish, then Theorem 2 give us a surjective map $H_2(B(A)) \longrightarrow H_2(\mathrm{SL}_2(A))$.

Theorem 3. *If $|k|$ is not exceptional, then there is a surjective map*

$$H_2(1 + \mathfrak{m}_A, \mathbb{Z}) \longrightarrow H_2(\mathrm{SL}_2(A), \mathbb{Z}).$$

In particular, when $1 + \mathfrak{m}_A$ is cylic, as in the case of fields or $\mathbb{Z}_{p^r}$, we obtain the classical result $H_2(\mathrm{SL}_2(A)) = 0$. The other cases and the study of the third homology will appear in a paper in preparation by the authors.

Other applications

In [2], Pérez extends the Bloch-Wigner sequence to local rings. We present two cases:

Theorem 4. *Let A be a local domain of characteristic 2, where its residue field has more than 64 elements. Then we have the refined Bloch-Wigner exact sequence*

$$0 \to \mathrm{Tor}^{\mathbb{Z}}_1(\mu(A), \mu(A)) \to H_3\left(SL_2(A), \mathbb{Z}\right) \to \mathcal{RB}(A) \to 0.$$

Theorem 5. *Let A be a conmutative local ring of characteristic $p \geq 3$, such that $H_i(\mathrm{T}(A), \mathbb{Z}) = H_i(\mathrm{B}(A), \mathbb{Z})$ for $i = 2, 3$. Then we have the exact sequence*

$$H_3\left(SM_2(A), \mathbb{Z}\right) \to H_3\left(SL_2(A), \mathbb{Z}\right) \to \mathcal{RB}(A) \to 0.$$

Other results obtained using these methods are:

- Mirzaii and Pérez studied the low dimensional homologies of projective linear and projective special linear groups [3, 4]. For that, appropiate spectral sequences are constructed. They also show that $K_i(A)[\frac{1}{2}] \sim H_i(\mathrm{PGL}_2(A), \mathbb{Z}[\frac{1}{2}])$ for $i = 2, 3$ and mild conditions on A. In particular, the abelianization of this groups has been computed.
- Mirzaii and Pérez generalized the construction presented here to study the low dimensional homology of the elementary group for general commutative rings [5]. As an application they prove a Bloch-Wigner sequence for semilocal rings with mild restrictions.

Acknowledgments: This study was financed in part by the Coordenação de Aperfeiçoamento de Pessoal de Nível Superior – Brasil (CAPES) – Finance Code 001.

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